Periodic Fever, Immunodeficiency, and Thrombocytopenia Syndrome (PFIT): Comprehensive Disease Report

Disease: Periodic Fever, Immunodeficiency, and Thrombocytopenia Syndrome (PFIT) **MONDO:** MONDO:0007883 · **Category:** Mendelian (autosomal recessive) **Causal gene:** *WDR1* (encoding Aip1 / actin-interacting protein 1)

Summary

Periodic Fever, Immunodeficiency, and Thrombocytopenia syndrome (PFIT) is an ultra-rare, autosomal-recessive **monogenic autoinflammatory "actinopathy"** caused by **biallelic hypomorphic missense mutations in *WDR1***, the gene encoding **actin-interacting protein 1 (Aip1)**. Aip1 is a WD40 β-propeller cofactor that cooperates with cofilin/ADF to accelerate depolymerization and turnover of actin filaments. Because the actin cytoskeleton is indispensable to leukocyte motility, immune-synapse formation, and megakaryocyte/platelet biogenesis, partial loss of Aip1 function produces a distinctive triad of **recurrent/periodic fevers (autoinflammation), immunodeficiency, and thrombocytopenia**, accompanied by severe stomatitis, oral stenosis, skin ulceration, and impaired wound healing. The condition typically presents in the neonatal period or early childhood and, untreated, can be fatal.

Mechanistically, PFIT is defined by two coupled abnormalities. First, loss of Aip1 **elevates neutrophil F-actin roughly four-fold** and impairs chemotaxis, spreading, and polarization while paradoxically preserving microbial killing and increasing oxidative burst. Second, the disease drives an **IL-18–dominant autoinflammatory state**: patients have high serum IL-18 (without a matching rise in IL-1β or IL-18–binding protein) and increased caspase-1 cleavage in monocytes, and mutant WDR1 protein aggregates that sequester pyrin, plausibly precipitating inflammasome assembly. Beyond myeloid cells, WDR1 deficiency causes lymphoid immunodeficiency — aberrant T-cell activation at the immunologic synapse and severe B-cell abnormalities (B lymphopenia, loss of switched memory B cells) — and macrothrombocytopenia from defective megakaryocyte maturation.

WDR1 is among the most **loss-of-function–intolerant genes in the human genome** (gnomAD pLI ≈ 1.0), which explains why every reported human patient carries **hypomorphic missense** alleles rather than complete-null variants: in mice, severe *Wdr1* loss of function is embryonic-lethal, whereas hypomorphic alleles are viable and reproduce the human phenotype (macrothrombocytopenia plus neutrophil-driven autoinflammation). The mechanism is deeply evolutionarily conserved from yeast (*AIP1*), *C. elegans* (*unc-78*), and plants to mouse *Wdr1*. **Allogeneic hematopoietic stem cell transplantation (HSCT) is the definitive, corrective treatment**, having reversed the immunologic defect in transplanted patients; diagnosis is genetic (WES/WGS or targeted autoinflammatory/immunodeficiency panels). This report consolidates eight confirmed findings across three primary human cohorts (total ~12 patients) and multiple mechanistic and model-organism studies.

Key Findings

Finding 1 — PFIT is caused by biallelic (autosomal recessive) missense mutations in *WDR1* (encoding Aip1)

PFIT was defined independently by two 2016–2017 index reports. Kuhns et al. (2016) identified **biallelic *WDR1* mutations affecting distinct antiparallel β-strands of Aip1** in four children across three families; heterozygous relatives were clinically normal, establishing autosomal-recessive inheritance. Standing et al. (2017) reported a **homozygous missense *WDR1* mutation in two siblings** with the periodic-fever/immunodeficiency/thrombocytopenia triad, coining the PFIT acronym. WDR1 encodes actin-interacting protein 1 (Aip1), a cofactor that accelerates cofilin-mediated actin filament depolymerization.

> "We report a homozygous missense mutation in WDR1 in two siblings causing periodic fevers with immunodeficiency and thrombocytopenia." — PMID: 27994071
>
> "Biallelic mutations in WDR1 affecting distinct antiparallel β-strands of Aip1 were identified in all patients... Heterozygous mutations in clinically normal relatives confirmed that WDR1 deficiency is autosomal recessive." — PMID: 27557945

Gene identifiers: *WDR1* — HGNC:12754, NCBI Gene 9948, Ensembl ENSG00000071127, UniProt O75083 (Aip1), gene OMIM *604734, cytoband 4p16.1 (GRCh38 chr4:~10.07–10.12 Mb, minus strand).

Finding 2 — Loss of Aip1 elevates F-actin and produces distinctive neutrophil morphologic and motility defects

In patient neutrophils, **F-actin was elevated ~4-fold**, and **chemotaxis and chemokinesis were markedly impaired**. Cells showed a distinctive morphology — herniation of nuclear lobes and agranular cytosolic regions — with impaired spreading on glass and defective polarization. Notably, **staphylococcal killing was preserved and oxidative burst was paradoxically increased** at baseline and on stimulation, indicating a specific defect in actin-dependent motility rather than a global neutrophil failure. Pfajfer et al. (2018) additionally documented defective adhesion and motility of neutrophils and monocytes. The biochemical basis is that Aip1 destabilizes cofilin-saturated actin filaments by severing them and accelerating monomer dissociation from both barbed and pointed ends; loss of this activity impairs filament turnover and lets F-actin accumulate.

> "Neutrophil F-actin was elevated fourfold, suggesting an abnormality in F-actin regulation." — PMID: 27557945
>
> "Chemotaxis and chemokinesis were markedly impaired, but staphylococcal killing was normal, and neutrophil oxidative burst was increased both basally and on stimulation." — PMID: 27557945
>
> "Aip1 also augments the monomer dissociation rate at both the barbed and pointed ends of actin." — PMID: 25448002

Finding 3 — PFIT autoinflammation is IL-18–dominant and driven by increased inflammasome/caspase-1 activity linked to pyrin

Standing et al. (2017) found PFIT patients had **high serum IL-18 without a corresponding rise in IL-18–binding protein or IL-1β**; patient cells secreted more IL-18 but not IL-1β in culture. **Increased caspase-1 cleavage in patient monocytes** indicated heightened inflammasome activity. In HEK293T co-transfection experiments, **mutant WDR1 protein formed aggregates**

that accumulated pyrin, potentially precipitating inflammasome assembly. This extends the *Wdr1*-mutant mouse model, in which autoinflammatory disease is bone-marrow–derived, nonlymphoid, and characterized by massive neutrophil infiltration.

> "Patients had high serum levels of IL-18, without a corresponding increase in IL-18-binding protein or IL-1β, and their cells also secreted more IL-18 but not IL-1β in culture." — PMID: 27994071
>
> "We found increased caspase-1 cleavage within patient monocytes indicative of increased inflammasome activity." — PMID: 27994071
>
> "Mutant protein formed aggregates that appeared to accumulate pyrin; this could potentially precipitate inflammasome assembly." — PMID: 27994071
>
> "Autoinflammatory disease, which is bone marrow-derived yet nonlymphoid in origin, is characterized by a massive infiltration of neutrophils into inflammatory lesions." — PMID: 17515402

Finding 4 — WDR1 deficiency also causes lymphoid immunodeficiency and macrothrombocytopenia

Pfajfer et al. (2018) identified novel homozygous/compound-heterozygous *WDR1* missense mutations in **6 patients from 3 kindreds** presenting with respiratory tract infections, skin ulceration, and stomatitis. Beyond myeloid defects, WDR1 deficiency caused **aberrant T-cell activation** — atypical actin accumulation at the immunologic synapse, reduced calcium flux, mildly impaired proliferation, and selective loss of follicular helper T cells — and **severe B-cell abnormalities**: peripheral B-cell lymphopenia, paucity of bone-marrow B-cell progenitors, and lack of switched memory B cells. Thrombocytopenia parallels the macrothrombocytopenia of hypomorphic *Wdr1* mice, which arises from megakaryocyte maturation defects causing failure of normal platelet shedding.

> "we identified novel homozygous and compound heterozygous WDR1 missense mutations in 6 patients belonging to 3 kindreds who presented with respiratory tract infections, skin ulceration, and stomatitis." — PMID: 29751004
>
> "WDR1 deficiency was associated with even more severe abnormalities of the B-cell compartment, including peripheral B-cell lymphopenia, paucity of B-cell progenitors in the bone marrow, lack of switched memory B cells." — PMID: 29751004

> "peripheral T cells from the patients accumulated atypical actin structures at the immunologic synapse and displayed reduced calcium flux and mildly impaired proliferation on T-cell receptor stimulation." — PMID: 29751004
>
> "Macrothrombocytopenia is the result of megakaryocyte maturation defects, which lead to a failure of normal platelet shedding." — PMID: 17515402

Finding 5 — PFIT belongs to the autoinflammatory "actinopathies"; onset is neonatal/early-childhood, and allogeneic HSCT is definitive

Kuhns et al. (2016) reported that **allogeneic stem cell transplantation corrected the immunologic defect** in one patient, whereas untreated patients had severe outcomes including oral stenosis and death. Reviews of autoinflammatory actinopathies (Mertz et al. 2024) classify WDR1 deficiency within this emerging autoinflammatory-disease subgroup, which typically manifests in the neonatal period and combines primary immunodeficiency, cytopenia (especially thrombocytopenia), and autoinflammation affecting skin and digestive system — often requiring allogeneic marrow transplantation.

> "Allogeneic stem cell transplantation corrected the immunologic defect in 1 patient." — PMID: 27557945
>
> "These diseases typically manifest in the neonatal period and variably combine a primary immunodeficiency of varying severity, cytopenia (particularly thrombocytopenia), autoinflammatory manifestations primarily affecting the skin and digestive system." — PMID: 39644982
>
> "In most cases, the severity of the conditions necessitates allogeneic marrow transplantation as a treatment." — PMID: 39644982

Finding 6 — *WDR1* is extremely loss-of-function-intolerant, so human PFIT arises only from hypomorphic biallelic missense variants

gnomAD constraint metrics place *WDR1* among the most LoF-intolerant genes: **pLI = 0.9999**, LOEUF (oe_lof upper) = 0.42, observed/expected LoF = 0.29, lof_z = 5.09. Consistent with this, **all reported human PFIT patients carry biallelic missense (hypomorphic) variants** — Kuhns 2016 (distinct antiparallel β-strands of Aip1), Standing 2017 (homozygous missense), Pfajfer

2018 (homozygous and compound-heterozygous missense). This mirrors the *Wdr1* mouse allelic series, in which severe loss of function is embryonic-lethal but hypomorphic alleles are viable and produce autoinflammation plus macrothrombocytopenia.

> "Biallelic mutations in WDR1 affecting distinct antiparallel β-strands of Aip1 were identified in all patients." — PMID: 27557945
>
> "While severe loss of function at the Wdr1 locus causes embryonic lethality, macrothrombocytopenia and autoinflammatory disease develop in mice carrying hypomorphic alleles." — PMID: 17515402

Finding 7 — Wdr1/Aip1 function and disease are evolutionarily conserved; mouse models recapitulate F-actin accumulation and tissue pathology

Two complementary mouse models establish causality and conserved mechanism. **(1)** Hypomorphic *Wdr1* mice (Kile et al. 2007) develop macrothrombocytopenia (megakaryocyte maturation defect) plus bone-marrow–derived, nonlymphoid autoinflammatory disease with massive neutrophil infiltration; severe LoF is embryonic-lethal. **(2)** A cardiomyocyte-specific *Wdr1* conditional knockout (Yuan et al. 2014) died by postnatal day 24 with cardiac hypertrophy, impaired left-ventricular contraction, prolonged QT, and progressive **F-actin accumulation within myofibrils**, with ectopic cofilin colocalizing at the aggregates. Aip1 orthologs are deeply conserved: *S. cerevisiae* AIP1, *C. elegans* unc-78, plant AIP1 (rice/*Arabidopsis*), and mouse *Wdr1* (NCBI Gene 22388), all promoting cofilin/ADF-mediated actin disassembly.

> "Actin filament (F-actin) accumulations began at P10 and became prominent at P12 in the myocardium of cKO mice… Ectopic cofilin colocalized with F-actin aggregates." — PMID: 24840128
>
> "These studies establish an essential requirement for Wdr1 in megakaryocytes and neutrophils, indicating that cofilin-mediated actin dynamics are critically important to the development and function of both cell types." — PMID: 17515402

Finding 8 — PFIT clinical phenotype spectrum and suggested HPO annotations

Across the three reported cohorts (Kuhns 2016 n=4; Standing 2017 n=2; Pfajfer 2018 n=6), the recurrent core features are: recurrent/periodic fevers, immunodeficiency with recurrent bacterial and respiratory infections, thrombocytopenia/bleeding tendency (including macrothrombocytopenia), mild neutropenia, severe stomatitis/recurrent oral ulceration progressing to oral stenosis, skin ulceration, and impaired wound healing. Laboratory hallmarks include markedly elevated serum IL-18 (with normal/low IL-1β and IL-18BP), elevated neutrophil F-actin (4-fold), distinctive neutrophil nuclear herniation with agranular cytosolic regions, B lymphopenia, and lack of switched memory B cells.

> "we identified 4 children with recurrent infections and varying clinical manifestations including mild neutropenia, impaired wound healing, severe stomatitis with oral stenosis, and death." — PMID: 27557945

Phenotype	Suggested HPO term	Frequency / notes
Recurrent/periodic fever (autoinflammation)	HP:0001954 (recurrent fever)	Core; episodic flares
Recurrent respiratory infections	HP:0002205	Common in Pfajfer cohort
Recurrent bacterial infections	HP:0002718	Core immunodeficiency feature
Thrombocytopenia	HP:0001873	Core
Macrothrombocytopenia (large platelets)	HP:0040314	Parallels mouse model
Neutropenia (mild)	HP:0001875	Variable
Recurrent oral ulceration / stomatitis	HP:0010280 (oral ulcer)	Severe; can progress
Oral stenosis	HP:0011099 (functional/ anatomic stenosis)	Severe untreated cases
Skin ulceration	HP:0200042	Pfajfer cohort
Poor/impaired wound healing	HP:0001058	Core
Elevated serum IL-18	(laboratory biomarker)	Diagnostic hallmark

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic hypomorphic missense mutation in *WDR1*** (4p16.1) **leads to** a partially functional Aip1 β-propeller protein (complete-null alleles are embryonic-lethal, so only hypomorphs survive).
2. Reduced Aip1 activity **results in** failure to sever and disassemble cofilin/ADF-bound actin filaments (Aip1 normally accelerates monomer dissociation from both filament ends).
3. Impaired filament turnover **leads to** pathological **accumulation of F-actin** (~4-fold in neutrophils) and aberrant actin structures in hematopoietic and other cells.

4. Elevated/dysregulated F-actin **branches** into three effector arms:
5. **Arm A — Myeloid/innate:** impaired neutrophil chemotaxis, spreading, and polarization (motility defect) with preserved killing and increased oxidative burst → recurrent infections + tissue neutrophil infiltration.
6. **Arm B — Autoinflammation:** mutant WDR1 aggregates sequester **pyrin**, and cytoskeletal dysregulation acts as a homeostasis-altering signal → **inflammasome/caspase-1 activation** → **IL-18–dominant** cytokine release (without IL-1β) → periodic fevers, cutaneous-digestive inflammation.
7. **Arm C — Lymphoid + platelets:** defective immune-synapse actin dynamics → T-cell activation defects + B-cell developmental failure (immunodeficiency); megakaryocyte maturation failure → defective platelet shedding → **macrothrombocytopenia/bleeding**.
8. The combined arms **result in** the clinical triad — **periodic fever** + **immunodeficiency** + **thrombocytopenia** — plus stomatitis, oral stenosis, skin ulceration, and poor wound healing; untreated disease can be fatal.
9. Because the defect is intrinsic to bone-marrow–derived cells, **allogeneic HSCT replaces the defective hematopoietic compartment and corrects the disease** (definitive treatment).

(Step 4B's pyrin/inflammasome link is mechanistically supported by co-transfection aggregation data and elevated caspase-1 cleavage but the precise molecular trigger connecting F-actin dysregulation to inflammasome assembly remains partly inferred.)

```
WDR1 missense (hypomorphic, biallelic)
       │
       ▼
Reduced Aip1 activity ──► cofilin-bound filaments not disassembled
       │
       ▼
F-actin accumulation (~4× in neutrophils)
       │
 ┌─────┼─────────────────────────┐
 ▼     ▼                         ▼
Neutrophil   Inflammasome/pyrin   Lymphoid + megakaryocyte
motility     → caspase-1          synapse/maturation defects
defect       → IL-18↑ (IL-1β–)    │
 │              │                 ├─ T/B-cell immunodeficiency
 ▼              ▼                 └─ macrothrombocytopenia
Recurrent    Periodic fever,          (bleeding)
infections   skin/digestive
             autoinflammation
       │
       ▼
  PFIT clinical triad → (untreated) death │ (HSCT) correction
```

Upstream vs downstream

- **Upstream (initiating):** *WDR1* mutation → Aip1 hypofunction → defective cofilin-mediated actin disassembly (molecular lesion).
- **Central node:** F-actin accumulation / disrupted actin turnover.
- **Downstream (effector):** neutrophil/lymphocyte dysfunction, inflammasome-driven IL-18, megakaryocyte failure → the clinical phenotype.

Cell types and processes (suggested ontology terms)

- **Cells (CL):** neutrophil (CL:0000775), monocyte (CL:0000576), T cell (CL:0000084), B cell (CL:0000236), megakaryocyte (CL:0000556), platelet (CL:0000233).
- **Biological processes (GO):** actin filament depolymerization (GO:0030042), regulation of actin cytoskeleton organization (GO:0032956), neutrophil chemotaxis (GO:0030593), inflammasome complex assembly (GO:0044546), interleukin-18 production (GO:0032620), megakaryocyte differentiation (GO:0030219), immunological synapse formation (GO:0001771).
- **Cellular components (GO CC):** cortical actin cytoskeleton (GO:0030864), lamellipodium (GO:0030027), actin filament (GO:0005884).

- **Anatomy (UBERON):** bone marrow (UBERON:0002371), blood (UBERON:0000178), oral mucosa (UBERON:0003729), skin (UBERON:0002097).
- **Chemical entities (CHEBI):** interleukin-18 (protein), reactive oxygen species (CHEBI:26523).

Report by Template Section

1. Disease Information

PFIT is an ultra-rare Mendelian autoinflammatory immunodeficiency defined by the triad of periodic fever, immunodeficiency, and thrombocytopenia, caused by *WDR1* mutations. **Identifiers:** MONDO:0007883; gene OMIM *604734 (*WDR1*). MONDO:0007883 is historically cross-referenced to OMIM 150550 ("lazy leukocyte syndrome"), an older descriptive label for a neutrophil-motility disorder — a mapping caveat worth flagging for curators, who should reconcile it against the specific WDR1-PFIT phenotype entry. **Synonyms:** WDR1 deficiency; Aip1 deficiency; autoinflammatory PFIT; a member of the "autoinflammatory actinopathies." Information is derived from **aggregated disease-level resources plus small primary case series** (≈12 patients across 3 kindreds/cohorts), not EHR-scale data.

2. Etiology

Causal factor: monogenic — biallelic hypomorphic **missense** mutations in *WDR1*. **Genetic risk:** requires two pathogenic alleles (AR); consanguinity increases risk (homozygous cases reported). No environmental cause; infections are downstream consequences of immunodeficiency rather than causes. **Protective factors:** none established. **Gene–environment interaction:** infectious exposures likely precipitate flares and morbidity in an immunodeficient host, but no formal GxE data exist.

3. Phenotypes

See Finding 8 table. Onset is **neonatal/early childhood**; severity is variable but often severe; course is chronic with **episodic autoinflammatory flares**. Quality-of-life impact is substantial (recurrent infections, painful stomatitis/oral stenosis limiting feeding, bleeding tendency); formal QoL instruments (EQ-5D/SF-36) have not been reported for this ultra-rare disease.

4. Genetic / Molecular Information

Causal gene: *WDR1* (HGNC:12754; NCBI Gene 9948; Ensembl ENSG00000071127; UniProt O75083; 4p16.1). **Variant class:** missense (hypomorphic) — pathogenic/likely pathogenic per ACMG in reported families; affecting the Aip1 β-propeller (distinct antiparallel β-strands). **Functional consequence:** partial loss of function (impaired cofilin-dependent actin disassembly). **Constraint:** pLI ≈ 1.0, LOEUF 0.42 — strong LoF intolerance, so null alleles are not observed in surviving patients. **Origin:** germline. **Modifier genes / epigenetics / chromosomal abnormalities:** none established.

5. Environmental Information

No toxic, occupational, or lifestyle etiologic factors. **Infectious agents** (e.g., *Staphylococcus*, respiratory pathogens) act as complicating/triggering factors in the immunodeficient host, not as primary causes.

6. Mechanism / Pathophysiology

See the ordered causal chain and diagram above. Core pathway: **cofilin/ADF–Aip1 actin-disassembly axis** → F-actin dysregulation → branching myeloid, inflammasome (IL-18/pyrin/caspase-1), and lymphoid/megakaryocyte effector arms. Molecular profiling to date is limited to targeted cytokine (IL-18) and cell-biological (F-actin, synapse) assays plus HEK293T co-transfection; no large-scale transcriptomic/proteomic/metabolomic PFIT datasets are published.

7. Anatomical Structures Affected

Primary: bone marrow / hematopoietic system (neutrophils, monocytes, T and B lymphocytes, megakaryocytes/platelets). **Secondary/target tissues:** oral mucosa (stomatitis, oral stenosis), skin (ulceration, poor healing), respiratory tract (infections). Involvement is **systemic/bilateral** (hematologic). Subcellular: cortical actin cytoskeleton, lamellipodium; F-actin aggregates.

8. Temporal Development

Onset: neonatal/early childhood. **Pattern:** chronic disease with episodic (periodic) autoinflammatory fever flares superimposed on persistent immunodeficiency and thrombocytopenia. **Progression:** can be severe/progressive if untreated (oral stenosis, fatal outcomes). **Remission:** treatment-induced (HSCT corrective); no reliable spontaneous remission. Critical window: early diagnosis enables transplantation before irreversible complications.

9. Inheritance and Population

Inheritance: autosomal recessive. **Penetrance:** appears high/complete in biallelic carriers; heterozygotes are unaffected. **Expressivity:** variable (severity and organ emphasis differ across families). **Consanguinity:** contributory (homozygous cases). **Epidemiology:** ultra-rare — only ~12 reported patients across 3 kindreds/cohorts; true prevalence/incidence unknown. No founder effect established for *WDR1* PFIT; no strong sex bias reported.

10. Diagnostics

Genetic diagnosis is definitive: WES/WGS or targeted autoinflammatory/immunodeficiency gene panels identifying biallelic *WDR1* missense variants. **Supportive labs/biomarkers:** markedly elevated **serum IL-18** (with normal/low IL-1β and IL-18BP), thrombocytopenia with large platelets on smear, mild neutropenia, elevated inflammatory markers, **4-fold elevated neutrophil F-actin**, distinctive neutrophil nuclear herniation/agranular cytosol, B lymphopenia and absent switched memory B cells on immunophenotyping. **Differential diagnosis:** other autoinflammatory actinopathies (e.g., ARPC1B deficiency, MKL1, ACTB/Baraitser-Winter), other periodic fever syndromes (FMF, HIDS), and other inherited macrothrombocytopenias/immunodeficiencies. **Screening:** cascade genetic testing of at-risk relatives; carrier testing in families.

11. Outcome / Prognosis

Untreated: poor — recurrent infections, bleeding, severe stomatitis/oral stenosis, and reported **death**. **With HSCT:** immunologic defect corrected; substantially improved prognosis. Prognostic factors include age at diagnosis, severity of infections/bleeding, and access to transplant.

12. Treatment

Definitive: allogeneic hematopoietic stem cell transplantation (HSCT) — corrective, replaces the defective bone-marrow compartment (NCIT: Allogeneic Hematopoietic Stem Cell Transplantation, C15393). **Supportive/bridging:** antimicrobial prophylaxis and treatment of infections; management of bleeding/thrombocytopenia; targeting the IL-18/inflammasome autoinflammatory axis is mechanistically rational (IL-18/IL-1 pathway modulation, anti-inflammatory agents) but not established as curative. Wound and oral care for stomatitis/ulceration. **Pharmacogenomics:** not applicable/established.

13. Prevention

No primary prevention (monogenic). **Genetic counseling** for AR recurrence risk (25% per pregnancy for carrier couples), carrier and prenatal/preimplantation testing in known families, and **cascade screening**. Tertiary prevention centers on early transplant and infection prophylaxis to avert complications.

14. Other Species / Natural Disease

Aip1/Wdr1 is deeply conserved: mouse *Wdr1* (NCBI Gene 22388), *S. cerevisiae AIP1*, *C. elegans unc-78*, plant AIP1 (rice *OsAIP1*, *Arabidopsis*). No naturally occurring companion-animal PFIT disease is catalogued; relevance is via engineered models. Not zoonotic.

15. Model Organisms

Mouse (*Mus musculus*) is the principal model. **Hypomorphic *Wdr1* mice** (Kile 2007) recapitulate macrothrombocytopenia and neutrophil-driven autoinflammation; **severe LoF is embryonic-lethal** (allelic-series dose dependence mirroring human missense-only genotypes). **Cardiomyocyte-specific *Wdr1* cKO** (Yuan 2014) demonstrates in-vivo F-actin accumulation and tissue pathology. **In vitro:** HEK293T co-transfection shows mutant WDR1 aggregation and pyrin accumulation; patient-derived neutrophils/T/B cells provide cellular assays. Invertebrate/yeast orthologs (*unc-78*, *AIP1*) support mechanistic conservation. **Limitations:** cardiomyocyte cKO models tissue-specific actin pathology rather than the hematologic/autoinflammatory human disease; small human cohorts limit genotype–phenotype resolution.

Evidence Base

PMID	Title (abbrev.)	Role / contribution	Evidence type
27994071	Autoinflammatory PFIT caused by WDR1 mutation	Index report; defines PFIT triad; IL-18 dominance; pyrin/caspase-1	Human clinical
27557945	Cytoskeletal abnormalities & neutrophil dysfunction in WDR1 deficiency	4-patient cohort; 4× F-actin; AR inheritance; HSCT corrective	Human clinical
29751004	WDR1 mutations lead to aberrant lymphoid immunity	6-patient cohort; T-synapse defect; B-cell failure; mucocutaneous phenotype	Human clinical
17515402	Aip1/Wdr1 mutations cause autoinflammation & macrothrombocytopenia	Mouse allelic series; LoF lethality vs hypomorph viability	Model organism
24840128	Cardiomyocyte-specific Wdr1 knockout	In-vivo F-actin accumulation; ectopic cofilin colocalization	Model organism
25448002	Aip1 destabilizes cofilin-saturated filaments	Biochemical mechanism of Aip1 disassembly activity	In vitro
39644982	Monogenic autoinflammatory actinopathies review	Classification; neonatal onset; HSCT as mainstay	Review
37596178	Actinopathy-associated autoinflammatory diseases review	Diagnostic approach; cutaneous-digestive autoinflammation	Review
33558442	Cytoskeletal proteins in immune diseases	Context: WDR1 among actin regulators in immunopathology	Review
32846417	Actinopathies as a PID category	Framework for immunologic actinopathies	Review
14742433 · 14680631 · 16421248	AIP1/cofilin biochemistry	Mechanistic basis of Aip1-cofilin cooperative disassembly	In vitro / yeast
23134061			

PMID	Title (abbrev.)	Role / contribution	Evidence type
	Rice OsAIP1 promotes actin turnover	Evolutionary conservation of Aip1 function	Plant model

Consistency: All three human cohorts converge on biallelic *WDR1* missense variants, actin dysregulation, and the PFIT triad, with independent replication of AR inheritance and HSCT correction. Mouse and biochemical studies provide congruent mechanistic support. No published study in the reviewed set contradicts the core model.

Limitations and Knowledge Gaps

- **Very small n (~12 patients):** prevalence, incidence, penetrance, and full expressivity are imprecise; genotype–phenotype correlation is limited.
- **Inflammasome trigger partly inferred:** the exact molecular link from F-actin dysregulation to pyrin/inflammasome activation rests on co-transfection aggregation data and caspase-1 cleavage — direct in-vivo mechanistic proof in patient tissue is incomplete.
- **No large-scale omics:** no published PFIT transcriptomic/proteomic/single-cell datasets to define cell-type-resolved mechanisms.
- **Therapeutic evidence beyond HSCT is anecdotal:** targeted IL-18/inflammasome blockade is rational but not validated in trials.
- **Identifier ambiguity:** MONDO:0007883's historical cross-reference to OMIM 150550 ("lazy leukocyte syndrome") should be verified against the specific WDR1-PFIT OMIM phenotype entry for curation accuracy.

Proposed Follow-up Experiments / Actions

1. **Establish an international PFIT/WDR1 registry** to capture natural history, penetrance, genotype–phenotype correlations, and HSCT outcomes.
2. **Patient-derived single-cell multi-omics** (scRNA-seq/CITE-seq of bone marrow and blood) to resolve cell-type-specific actin/inflammasome dysregulation and IL-18 sources.

3. **Mechanistic dissection of the F-actin → pyrin/inflammasome link** using patient monocytes and CRISPR-engineered *WDR1*-hypomorph cell lines/organoids; test whether restoring actin turnover normalizes IL-18.
4. **Preclinical trials of IL-18/IL-1 pathway blockade** (as a bridge to or adjunct with HSCT) in hypomorphic *Wdr1* mice and patient cells.
5. **Structure–function studies** mapping reported β-propeller missense variants onto Aip1 structure to predict residual activity and severity (variant-interpretation aid).
6. **Curate ontology annotations** (HPO frequencies, GO/CL/UBERON/CHEBI terms above) into the disease knowledge base and reconcile MONDO/OMIM cross-references.

Report compiled from 8 confirmed findings and 18 reviewed papers across 5 investigation iterations. Evidence types are labeled human clinical, model organism, in vitro, or review throughout.
